Estimating the caffeine still in the body from self-reported drinks

Literature review, model options, calibration, validation and implementation specification for the *bezKRATOMU* application

Technical report (model version 2: smoking)

8 November 2026; version 2: 12 November 2026

Abstract

The *bezKRATOMU* application lets users log caffeine in milligrams (energy drinks, coffee and their own presets). This report specifies how the application will estimate how much of that caffeine is still in the body at any moment, what it will be at the user's bedtime, and how that compares with a soft, literature-derived reference level for sleep. Caffeine is among the best-studied substances in clinical pharmacology: oral absorption is fast and complete (bioavailability ≈ 100 %, peak after 30–120 min, the same for coffee, cola and energy drinks), it distributes in total body water ($V \approx 0.6$–0.75 L/kg) and it is eliminated almost entirely by hepatic CYP1A2 with first-order kinetics in the usual dose range. Reported mean elimination half-lives in 25 healthy adult control groups from peer-reviewed studies have a median of 5.20 h (interquartile range 4.3–5.6 h); individuals range from about 2 to 10 h, mainly because of smoking, oral contraceptives, pregnancy, liver function, interacting drugs and genetics. We compare four model options and recommend the simplest adequate one: a one-compartment model with first-order absorption ($k_a = 4.42\,\mathrm{h}^{-1}$, fitted to the median reported time to peak of 49 min) and first-order elimination ($t_{1/2} = 5.20$ h, one population average as decided by the user). Against independent published peak and late concentrations, the model is unbiased in shape but under-predicts early peak concentrations by about 20 % (median ratio 0.81), as expected from the omitted distribution phase; this does not affect the displayed quantity (milligrams in the body). A sleep reference of 0.52 mg per kg body weight (36 mg at 70 kg) is derived from the 2023 meta-analysis of caffeine and sleep and agrees with a direct measurement. The estimate is a population average, not a measurement, and the report states its limits.

Version 2 (12 November 2026). For users who log cigarettes in the application, the elimination rate now follows their own smoking (Section 8). Caffeine clearance rises with the number of cigarettes in the last 24 h (fitted to 786 people, Tantcheva-Poór 1999: ×1.52 at 10 and ×1.69 at 20 cigarettes a day, i.e. a half-life of 3.4 and 3.1 h) and follows changes in smoking with the 38.6 h half-life measured after quitting (Faber 2004). Against seven independent smoker/non-smoker and quitting comparisons the model is within −14 to +23 %. Snus and nicotine replacement are not counted. Without logged cigarettes the model is unchanged.

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1 Purpose and decisions

The application already tracks caffeine entries (mg, time) per user. This model adds a card "*Kofein v těle*" analogous to the kratom card (*V těle teď*). The owner of the application decided (8 November 2026):

1. The main number is the **milligrams of caffeine still in the body**, with a secondary line in **cups of coffee** ("≈ X šálků kávy").
2. **One population-average half-life** for everyone; no adjustment for smoking, oral contraceptives or pregnancy (these are documented here as options for later, Section 4). *Changed in version 2 (12 November 2026):* the owner decided that a user's own cigarette log may adjust that user's own caffeine estimate (Section 8); contraceptives and pregnancy are still not used.
3. **Caffeine only**; its main metabolite paraxanthine is not shown (Section 3.3).
4. Sleep: a **projection at the user's bedtime** and a **soft reference level** from the literature, clearly labelled as such.
5. **Body weight** defaults to 70 kg and can be changed in the settings.
6. The model is specified now; deployment to the application comes later (Section 9).

As with the kratom model, the estimate is computed only from the user's own entries and shown only to them; it is an estimate for orientation, never a measurement and never advice on when to consume caffeine.

2 Literature search

Sources and queries. PubMed was searched through the NCBI E-utilities on 8 November 2026 with targeted queries (caffeine bioavailability, pharmacokinetics in healthy volunteers, dose dependence, population pharmacokinetics, absorption from coffee, cola and energy drinks, sleep, smoking, oral contraceptives, pregnancy, caffeine content of coffee) and one systematic query,

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(caffeine[MeSH Major Topic] OR caffeine[ti]) AND (pharmacokinetic*[tiab] OR half-life[tiab]
OR clearance[tiab]) AND (healthy[tiab] OR volunteers[tiab]) AND humans[mh] AND english[la]
NOT (infant*[ti] OR neonat*[ti] OR preterm[ti])
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which returned 208 records. Abstracts were screened; full texts were read where open access (Grzegorzewski 2021, Fendt 2021, White 2016).

Inclusion. Peer-reviewed original studies and systematic reviews/meta-analyses in humans, predominantly in clinical pharmacology journals (*Clin Pharmacol Ther*, *Br J Clin Pharmacol*, *Eur J Clin Pharmacol*, *J Clin Pharmacol*, *Clin Pharmacokinet*, *CPT Pharmacometrics Syst Pharmacol*, *Pharmacol Rev*) and leading sleep journals (*Sleep Med Rev*, *J Clin Sleep Med*). For the population half-life, only healthy adult *control* groups (or baseline/placebo phases) given caffeine by mouth were used; groups defined by smoking, oral

contraceptives, pregnancy, disease or an interacting drug were excluded. Neonatal studies were excluded. Conference abstracts, grey literature and non-reviewed web sources were not used.

Pooled data. The most comprehensive source, a systematic analysis of 141 publications (Grzegorzewski 2021), deposited its curated data in the PK-DB database; the database could not be reached during this work, so the study-level synthesis in Section 5 was rebuilt from the original abstracts. Its conclusions (clear, reproducible effects of smoking and oral contraceptives; no systematic assay effect) agree with ours.

3 What is known about caffeine pharmacokinetics

3.1 Absorption

Oral caffeine is absorbed rapidly and completely. In a randomised crossover study against intravenous dosing, the absolute bioavailability of an oral solution was 108 ± 4 %, i.e. complete, with the peak after 30 ± 8 min (Blanchard 1983). Complete availability was confirmed from 50 to 750 mg (Newton 1981). The vehicle matters little: peak saliva levels after 400 mg were the same from coffee and cola (42 and 39 min) and somewhat later from capsules (67 min; Liguori 1997), and in a crossover study of 160 mg from hot or cold coffee and a sugar-free energy drink, drunk in 2 or 20 minutes, the concentration–time profiles overlapped with no significant pairwise differences (White 2016). The model therefore uses one absorption rate for all drinks and treats each entry as taken at its logged time.

3.2 Distribution

Caffeine distributes in total body water. Reported volumes of distribution in healthy adults are 0.63–0.75 L/kg (Table 3); protein binding is low (unbound volume about 1.06 L/kg; Lelo 1986). Saliva concentrations are about 74 % of plasma (Newton 1981), which allows saliva studies to be used.

3.3 Metabolism and elimination

About 95 % of a dose is metabolised in the liver by CYP1A2 (Nehlig 2018); less than 2 % is excreted unchanged in urine (Newton 1981). Roughly 80 % becomes paraxanthine, which is itself a stimulant with a similar clearance and a half-life of about 3 h (Lelo 1986). Paraxanthine is not modelled (decision 3); showing it would roughly add a delayed, smaller copy of the caffeine curve.

Elimination is first-order in the usual intake range in most studies (dose-normalised profiles superimposable from 50 to 750 mg; Newton 1981; dose-independent kinetics at usual intakes, Bonati 1982), but several studies found clearance falling at higher single doses (250 vs. 500 mg, Kaplan 1997; 70 to 300 mg, Cheng 1990) and with repeated daily dosing (Denaro 1990). This is the main structural uncertainty and is discussed as option B in Section 4.

3.4 Why people differ

Half-lives in healthy adults range from about 2 to 10 h (2.7–9.9 h in ten young men, Blanchard 1983). The main documented causes are:

- **Smoking** induces CYP1A2: half-life about 3.0–3.5 h in smokers vs. 4.3–6.0 h in non-smokers (Parsons 1978; Seng 2009); after quitting, clearance falls by about 36 % with a half-time of 39 h (Faber 2004). Smokeless nicotine does not induce CYP1A2 (the inducers are combustion products).
- **Oral contraceptives** roughly halve clearance: 7.9 vs. 5.4 h (Abernethy 1985), 9.7 vs. 5.0 h (Haller 2004).
- **Pregnancy**: 8.3 h on average (range 3–16 h) vs. 3.4 h outside pregnancy (Knutti 1981).
- **Liver disease** and **CYP1A2-inhibiting drugs** (e.g. fluvoxamine: 5 to 31 h, Jeppesen 1996; ciprofloxacin: 4.5 to 9.7 h, Mahr 1992; cannabidiol: 5.4 to 10.9 h, Thai 2021).
- **Genetics**: in non-smokers not using contraceptives, 89 % of the variation in caffeine exposure was attributed to genetic effects in a twin study cited by Grzegorzewski 2021; see also Nehlig 2018.

A population model (Seng 2009, one-compartment, 59 men) quantified inter-individual variability as a coefficient of variation of 33 % for clearance, 15 % for volume and 51 % for the absorption rate; these values are used for the uncertainty bands below.

4 Model options

Table 1: Model options considered.

Option	For	Against
A. One compartment, first-order absorption and elimination (recommended)	Structure used by the population PK model of healthy adults (Seng 2009); closed-form sums of exponentials, trivial to run in the browser and in SQL; three well-supported parameters.	Misses the short distribution phase (early peak concentrations about 20 % too low); linear in dose.
B. Saturable (Michaelis–Menten) elimination	Reflects the slower elimination reported after large doses and repeated daily intake (Kaplan 1997; Cheng 1990; Denaro 1990).	Evidence is mixed (linear up to 750 mg in Newton 1981); $V_{\max}$ and K_m are poorly identified in the literature; no closed form, so the browser must integrate numerically.
C. Two-compartment or full PBPK (e.g. Fendt 2021)	Better early concentrations; can be personalised with demographics and CYP1A2 phenotype.	Many parameters with no gain for the displayed quantity (mg in the body); even a personalised PBPK model predicted only 66 % of values within 1.25-fold (Fendt 2021).
D. Option A with covariates (smoking from logged cigarettes, contraceptives, pregnancy)	Captures the largest known sources of variation; smoking could come from the app's own cigarette log, with the 39 h de-induction half-time of Faber 2004.	Declined for version 1 (decision 2); needs extra, sensitive settings. Recommended as the first extension; **smoking adopted in version 2** (Section 8).

Option A is adopted. Figure 6 shows what option D would change: for the same day of drinks, the caffeine left at bedtime ranges from roughly a third to more than twice the population estimate.

5 Model and parameters

5.1 Equations

For a dose D (mg) taken t hours ago, with complete bioavailability,

$$\text{gut}(t) = D\, e^{-k_a t}, \tag{1}$$

$$\text{body}(t) = D\, \frac{k_a}{k_a - k_e} \left(e^{-k_e t} - e^{-k_a t}\right), \tag{2}$$

$$C(t) = \frac{\text{body}(t)}{V \cdot W}, \tag{3}$$

where body(t) is the caffeine absorbed and not yet eliminated (the amount distributed in blood and tissues), V the volume of distribution per kg and W the body weight. Several entries add up (superposition). Entries older than 72 h contribute less than 10^{-4} of their dose and are ignored.

5.2 Elimination half-life

Table 2 lists the 25 control groups used. Study means range from 3.4 to 7.1 h; the median is 5.20 h (interquartile range 4.3–5.6 h), so $k_e = \ln 2/5.20 = 0.1333\,\text{h}^{-1}$. The median of study means is used rather than a weighted mean because group sizes are not reported in all abstracts and the median is robust to the few small studies. This population includes men and women; because smokers (shorter half-life) and contraceptive users (longer) were excluded, it is a value for an "average" adult rather than for any of those groups.

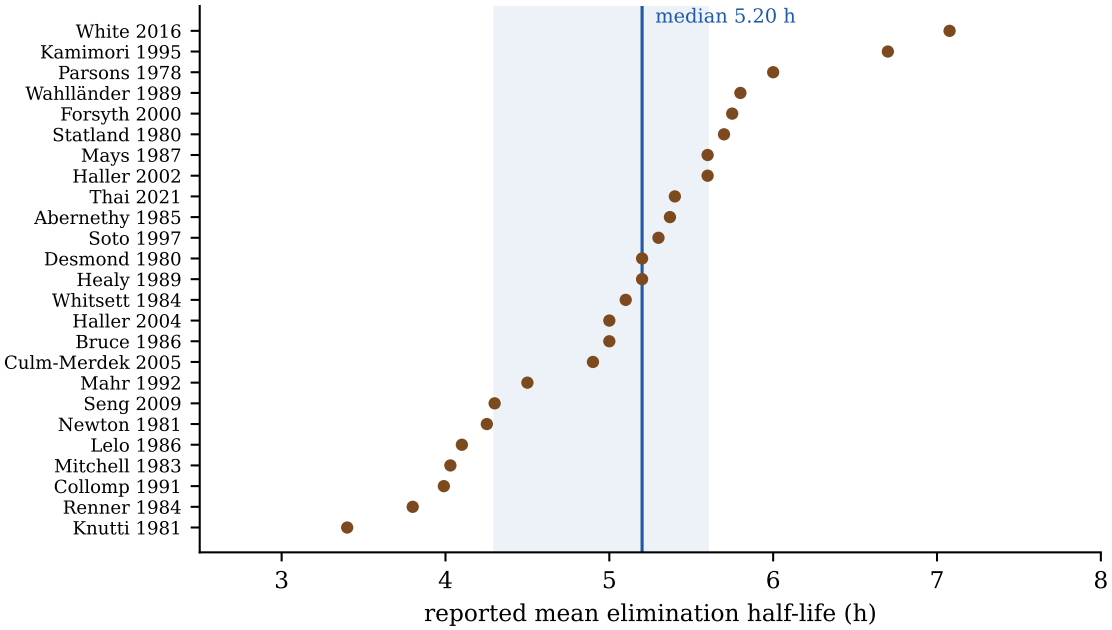


Figure 1: Mean elimination half-life reported for healthy adult control groups. Blue line: median; shaded: interquartile range.

Table 2: Half-life data (healthy adult control groups, oral caffeine).

Study	Journal	Group	$t_{1/2}$ (h)	PMID
Parsons 1978	*Clin Pharmacol Ther*	non-smokers	6.00	657717
Statland 1980	*Am J Clin Pathol*	healthy subjects	5.70	7361718
Desmond 1980	*Dig Dis Sci*	healthy controls	5.20	7371463
Knutti 1981	*Eur J Clin Pharmacol*	men and non-pregnant women (saliva)	3.40	7341280
Newton 1981	*Eur J Clin Pharmacol*	50 mg dose	4.25	7333346
Mitchell 1983	*J Lab Clin Med*	baseline	4.03	6854123
Renner 1984	*Hepatology*	normal volunteers	3.80	6420303
Whitsett 1984	*Am J Cardiol*	non-smokers	5.10	6702647
Abernethy 1985	*Eur J Clin Pharmacol*	women, no oral contraceptives	5.37	4029248
Lelo 1986	*Br J Clin Pharmacol*	men	4.10	3756065
Bruce 1986	*Br J Clin Pharmacol*	healthy (approximate)	5.00	3741730
Mays 1987	*Clin Pharmacol Ther*	baseline	5.60	3690940
Healy 1989	*Antimicrob Agents Chemother*	baseline	5.20	2729942
Wahlländer 1989	*Eur J Clin Pharmacol*	controls	5.80	2612543
Collomp 1991	*Eur J Clin Pharmacol*	at rest	3.99	2060565
Mahr 1992	*Clin Pharmacokinet*	placebo phase	4.50	1319876
Kamimori 1995	*Eur J Clin Pharmacol*	sea level	6.70	7589033
Soto 1997	*J Clin Pharm Ther*	baseline	5.30	9447474
Forsyth 2000	*Br J Clin Pharmacol*	healthy controls (saliva)	5.75	11012552
Haller 2002	*Clin Pharmacol Ther*	no oral contraceptives	5.60	12087345
Haller 2004	*Clin Pharmacol Ther*	men	5.00	15060505
Culm-Merdek 2005	*Br J Clin Pharmacol*	placebo phase	4.90	16236038
Seng 2009	*J Clin Pharm Ther*	non-smokers (population PK)	4.30	19125908
White 2016	*Clin Toxicol*	men and women, incl. OC users	7.08	27100333
Thai 2021	*Clin Pharmacol Drug Dev*	placebo phase	5.40	33951339

5.3 Volume of distribution and clearance

The median of the volumes in Table 3 is $V = 0.669\,\mathrm{L/kg}$. Together with k_e this implies a clearance of $1.49\,\mathrm{mL/min/kg}$, in the middle of the values reported for healthy non-smokers (Table 4), which supports the internal consistency of the two parameters.

Table 3: Volume of distribution.

Study	L/kg
Lelo 1986	0.68
Abernethy 1985	0.75
White 2016	0.66
Seng 2009 (42 L, CL 114 mL/min, assumed 65 kg)	0.65

Table 4: Reported clearance (model: 1.49).

Study	mL/min/kg
Lelo 1986	2.07
Abernethy 1985 (controls)	1.75
White 2016 (5 arms)	1.32
Wahlländer 1989 (controls)	1.34
Newton 1981	0.98
Faber 2004 (after quitting smoking)	1.53
Seng 2009 (114 mL/min, 65 kg)	1.75

5.4 Absorption rate

k_a was chosen so that the model peaks at the median reported time to peak after drinks and solutions (Table 5). Where a drink was sipped over several minutes (White 2016), time is counted from the middle of drinking. The median is 49 min, giving $k_a = 4.42\,\mathrm{h}^{-1}$ (absorption half-time about 9 min). Peak concentrations were *not* used for fitting, so they remain an independent check.

Table 5: Time to peak used for k_a.

Study, condition	$t_{\max}$ (min)
Blanchard 1983, solution 5 mg/kg	30
Liguori 1997, coffee 400 mg (saliva)	42
Liguori 1997, cola 400 mg (saliva)	39
Abernethy 1985, 162 mg (controls)	47
White 2016, hot coffee over 20 min	49
White 2016, cold coffee over 2 min	61
White 2016, cold coffee over 20 min	54
White 2016, energy drink over 2 min	69
White 2016, energy drink over 20 min	72

6 Validation

Table 6 compares predicted with observed concentrations from studies whose concentrations were not used to set any parameter. Landolt 1995 measured saliva; values were converted to plasma with the saliva/plasma ratio of 0.74 (Newton 1981), and 70 kg was assumed because weight was not reported.

Table 6: Independent validation (mg/L plasma).

Study	observed	predicted	ratio
Blanchard 1983 (solution, 5 mg/kg)	10.00	6.71	0.67
White 2016 hot coffee (160 mg, 74 kg)	3.74	2.90	0.78
White 2016 cold coffee 2 min	3.64	2.90	0.80
White 2016 cold coffee 20 min	3.56	2.90	0.81
White 2016 energy drink 2 min	3.36	2.90	0.86
White 2016 energy drink 20 min	3.14	2.90	0.92
Landolt 1995 peak (200 mg, saliva, 70 kg assumed)	4.46	3.80	0.85
Landolt 1995 at 23:00, 15.8 h later	0.79	0.53	0.68

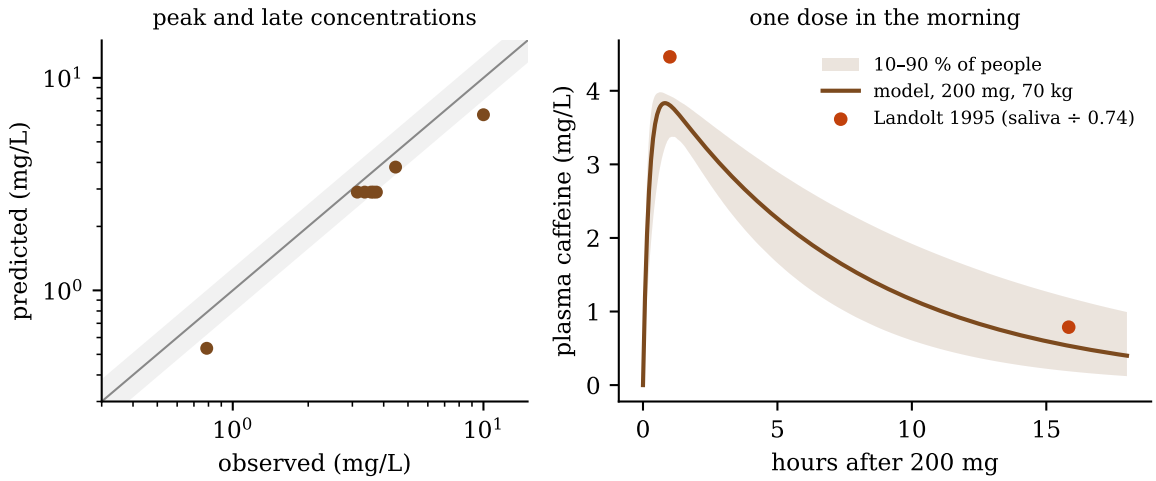


Figure 2: Left: predicted vs. observed concentrations (grey band: ±25 %). Right: model and 10–90 % population band after 200 mg, with the two measurements of Landolt 1995.

Findings. Prediction ratios range from 0.67 to 0.92 (median 0.81).

- *Peaks are about 20 % too low.* Studies report the mean of individual peaks, which is higher than the peak of a typical person; simulating the population variability accounts for only about 1 % of the gap (factor 1.01). The rest is the early distribution phase that a one-compartment model omits: just after absorption, caffeine has not yet spread through all body water, so blood levels are briefly higher than the volume at steady state implies. The oral solution taken fasting (Blanchard 1983) was also absorbed faster than drinks. Neither affects the amount in the body, which is what the application shows.
- *The late point (15.8 h) is under-predicted by about a third.* The two measurements of Landolt 1995 imply a half-life of about 5.9 h in that group, longer than the population median but inside its spread; the point lies inside the 10–90 % band (Figure 2).

The comparison supports the model shape and its order of magnitude; the individual uncertainty (next section) is larger than the model error.

7 What the application will show

7.1 Milligrams in the body

The main number is body(t) summed over the user's entries: caffeine absorbed and not yet eliminated. While a drink is still being absorbed (about the first hour), a small line can add "+ X mg se ještě vstřebává" (gut(t)), so that a freshly logged drink does not seem to disappear. Figure 3 shows one small and one large energy drink: 150 mg peaks at about 135 mg in the body after 49 min and is below 36 mg after about 10.9 h.

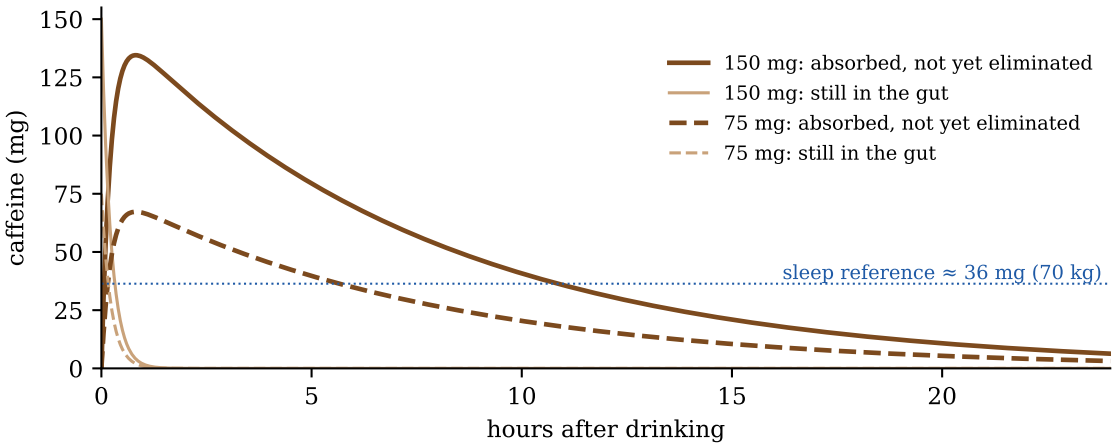


Figure 3: One large (150 mg) and one small (75 mg) energy drink.

7.2 Cups of coffee

A "cup of coffee" is not a standard amount: espressos bought in coffee shops contained 48–317 mg per serving (Ludwig 2014) and up to 322 mg (Crozier 2012). The sleep meta-analysis used 107 mg per 250 mL cup (Gardiner 2023). The application uses a round **100 mg = 1 šálek** and should say so in its explanation.

7.3 Bedtime and the sleep reference

Evidence. In a meta-analysis of 24 studies, caffeine reduced total sleep time by 45 min and sleep efficiency by 7 %, and lengthened sleep onset; to avoid a reduction in total sleep time, 107 mg (coffee) should be taken at least 8.8 h before bed and 217.5 mg (a pre-workout serving) at least 13.2 h before bed (Gardiner 2023). 400 mg taken 6 h before bed still disrupted sleep significantly in a randomised home study (Drake 2013). In a laboratory study, 200 mg at 07:10 still reduced sleep efficiency and total sleep time at 23:00, when the saliva level was 3 µmol/L (Landolt 1995).

Derivation. Run through the model, Gardiner's two limits correspond to 34 and 39 mg still in the body at bedtime, and Landolt's measured level to 37 mg at 70 kg: three independent routes that put the level from which sleep is measurably affected at roughly 35–40 mg for a 70 kg adult. Drake's condition corresponds to 185 mg, far above it, consistent with its clear effect (Figure 4). We take the mean of the two meta-analytic values, **36 mg at 70 kg**, and scale it with body weight as **0.52 mg/kg** (a plasma level of about 0.78 mg/L), because the effect depends on the concentration at the adenosine receptors, not on the amount.

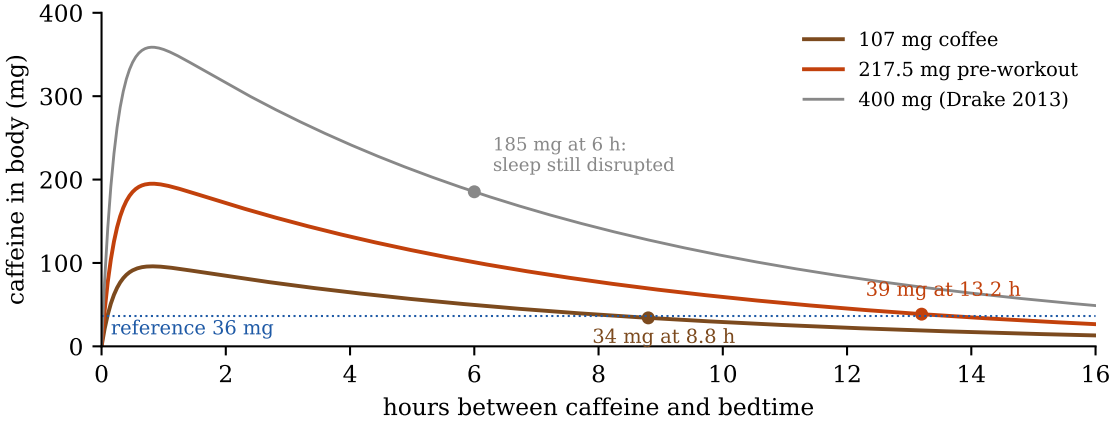


Figure 4: Caffeine in the body vs. hours before bedtime for the doses of Gardiner 2023 and Drake 2013. Dots: the published limits run through the model.

How to word it. This is a *soft* reference: it comes from average effects on total sleep time; deep-sleep EEG changes were seen even near this level (Landolt 1995), and sensitivity to caffeine varies with genetics (Nehlig 2018). The application should phrase it as "*kolem této hodnoty a výš kofein podle studií v průměru zkracuje spánek; citlivější lidé to můžou cítit i při méně*", never as a safe limit.

Bedtime projection. With a bedtime setting, the card shows body(t_{bed}) assuming no further caffeine, and when the estimate falls below the reference. For a typical day of three energy drinks (Figure 5), about 111 mg would remain at 23:00 for a typical person (10–90 %: 66–172 mg).

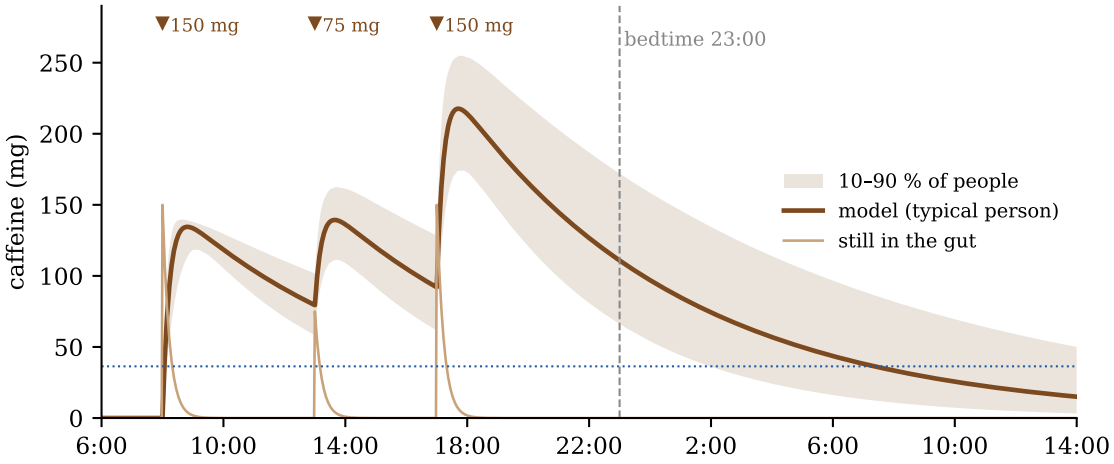


Figure 5: Large energy drink at 8:00, small at 13:00, large at 17:00. Dotted: sleep reference (70 kg). Band: 10–90 % of people (population variability of Seng 2009).

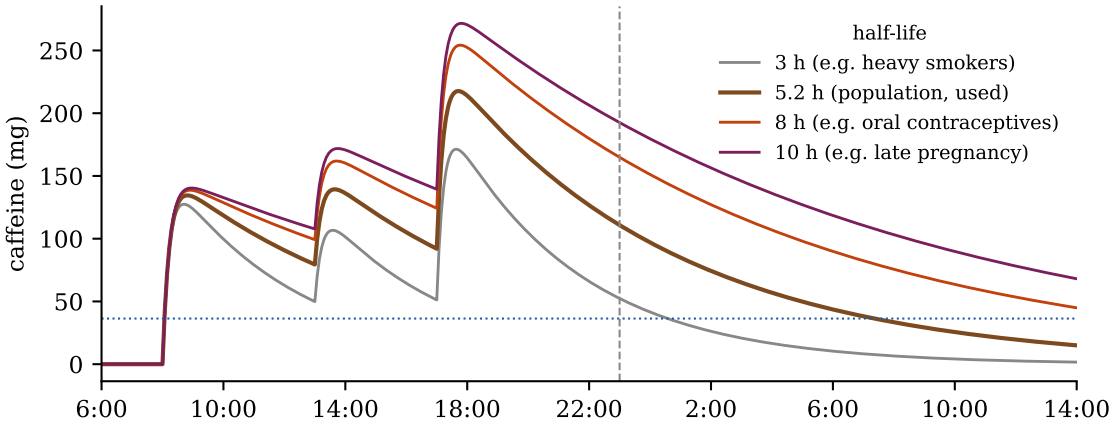


Figure 6: Same day with other half-lives (option D, not used in version 1).

8 Smoking (model version 2)

8.1 Question and search

The application also lets users log cigarettes and snus. Smoking is the largest well-documented source of variation in caffeine elimination (Section 3.4), so for users who log cigarettes the elimination rate is now made to follow their own smoking. Three questions had to be answered from the literature: how much smoking increases caffeine clearance as a function of the number of cigarettes, how fast the increase appears and disappears when smoking changes, and whether smokeless tobacco or nicotine alone have the same effect.

PubMed was searched on 12 November 2026 (E-utilities) for caffeine or CYP1A2 combined with smoking, cigarettes or tobacco and with clearance, half-life, metabolic ratio, induction or activity; for CYP1A2 or caffeine with cessation, abstinence or quitting; for CYP1A2 with cigarettes per day, dose dependence or

light smokers; for caffeine or CYP1A2 with snus, smokeless tobacco, nicotine replacement, e-cigarettes or vaping; and for meta-analyses (about 620 unique records). Titles were screened and abstracts of all human studies with caffeine (or another CYP1A2 probe) and smoking data were read. One letter on nicotine alone (Hukkanen 2011) could not be read in full.

8.2 What is known

- **Mechanism.** Polycyclic aromatic hydrocarbons in smoke induce hepatic CYP1A2 through the aryl hydrocarbon receptor; nicotine is not the inducer (Schaffer 2009; Anderson 2016). The volume of distribution is unchanged (Parsons 1978), so smoking acts on clearance only.
- **Size of the effect.** Smokers' caffeine clearance is 1.3–2.0 times that of non-smokers (Table 7); CYP1A2 activity ratios are similar (1.55, 194 smokers, Dobrinas 2011; 1.66–1.70, Vistisen 1991).
- **Dose-response.** In 786 healthy adults, caffeine clearance relative to non-smokers was 1.22, 1.47, 1.66 and 1.72 for 1–5, 6–10, 11–20 and >20 cigarettes a day (Tantcheva-Poór 1999). Hepatic CYP1A2 abundance derived independently for the same categories rises in the same way (Plowchalk 2012), and cigarettes per day and urinary cotinine correlate with CYP1A2 activity (Kalow 1991; Dobrinas 2011). The effect therefore rises steeply over the first ten cigarettes and then levels off.
- **Time course after quitting.** In 12 heavy smokers phenotyped repeatedly for 13 days after quitting, caffeine clearance fell monoexponentially by 36 % with a half-life of 38.6 h (95 % CI 27.4–54.4 h; Faber 2004), i.e. most of the change is over within a week. Consistent: 24-h caffeine AUC rose by 46 % after 4 days of abstinence (Brown 1988); clearance fell from 118 to 77 mL/min after 3 weeks (Joeres 1988). Switching to heated tobacco or e-cigarettes reverses the induction like quitting within about five days (van der Plas 2020; Uche 2026).
- **Starting or resuming smoking.** No time course of induction onset in humans was found. Case reports show that clozapine levels fall reproducibly when a patient resumes smoking 6 cigarettes a day (Ikawa 2026). In a turnover model the same enzyme degradation rate governs both directions, which is the assumption used here.
- **Smokeless tobacco and nicotine.** Theophylline (a CYP1A2 substrate) half-life in regular snuff or chewing-tobacco users who did not smoke was the same as in non-users (Parish 1989), and plasma, urine and saliva metabolomics showed altered caffeine metabolism in smokers but not in moist-snuff users (Liu 2019, industry-funded). Primary human hepatocytes exposed to smokeless-tobacco extracts *in vitro* did induce CYP1A2 mRNA (Lenich 2026); this has not been confirmed in people. Snus is therefore not counted.
- **Variability.** The induced increase varies widely between people (after quitting from no change to a 7.3-fold decrease; Dobrinas 2011), partly with CYP1A2 genotype.

8.3 Model

The elimination rate becomes time-dependent, $k_e(t) = k_e\, M(t)$, with k_e the population value of Section 5 (which excludes smokers) and $M(t) \geq 1$ a factor that follows the user's smoking:

$$r(t) = \text{cigarettes logged in } (t - 24\,\text{h},\, t], \tag{4}$$

$$M^*(r) = 1 + E_{\max}\,\frac{r}{EC_{50} + r}, \qquad E_{\max} = 1.05,\ EC_{50} = 10.3 \text{ cigarettes/day}, \tag{5}$$

$$\frac{dM}{dt} = k_{\deg}\left(M^*(r(t)) - M(t)\right), \qquad k_{\deg} = \ln 2/38.6\,\text{h}. \tag{6}$$

$E_{\max}$ and EC_{50} were fitted (least squares on the log scale) to the four Tantcheva-Poór categories, using midpoints and 25 for ">20" (Figure 7a). A user who has just started logging is assumed to have smoked before as in their first logged week (at least one day): M starts at $M^*(r_0)$ with r_0 that average, and the part of the 24-h window before the first logged cigarette counts at r_0. Cigarettes older than 28 days are ignored (2^{-17} of their effect is left). The projection ("if you have nothing more") keeps M at its current value: it assumes no more caffeine, not that the user stops smoking.

For each caffeine entry at time t_i,

$$\text{body}_i(t) = D_i\,\frac{k_a}{k_a - k_e M(t_i)}\left(e^{-\int_{t_i}^{t} k_e M(s)\,ds} - e^{-k_a (t - t_i)}\right), \tag{7}$$

i.e. elimination uses the exact integral of the varying rate, and absorption (which takes about an hour, while M changes over days) uses the rate at the time of the entry. Against a direct numerical solution of the differential equations in the test scenario below, the largest difference is 0.05 mg. With no cigarettes $M \equiv 1$ and everything is identical to version 1.

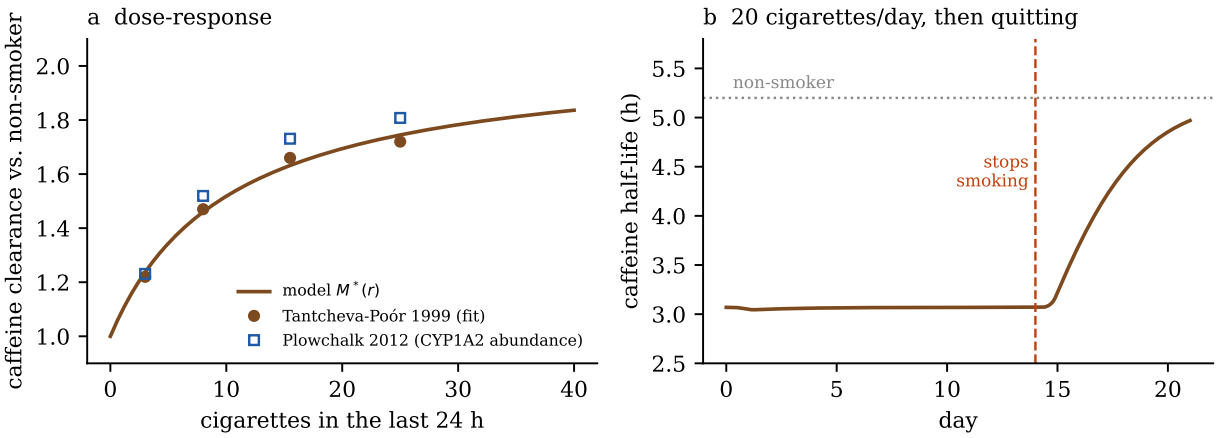


Figure 7: (a) Caffeine clearance relative to non-smokers by cigarettes in the last 24 h: model and data. (b) Caffeine half-life for a user who smokes 20 cigarettes a day for two weeks and then stops.

8.4 Validation

Table 7 compares the model with smoker/non-smoker and quitting studies that were not used for the fit. Where the number of cigarettes is not given, 20 a day is assumed. The model ratio is within −14 to +11 % of six of the seven observations and 23 % above the smallest within-subject effect (Benowitz 2003, 12 smokers; part of the induction may not yet have reversed after 7 days of breathing air, or the participants smoked less).

Table 7: Caffeine clearance ratio (smoking vs. non-smoking), observed and model.

Study	Design	Observed	Model	Model/obs.	Note
Parsons 1978	13 smokers vs 13 non-smokers	1.65	1.69	1.03	cigarettes/day not given; 20 assumed
Joeres 1988	8 smokers vs 15 non-smokers	1.78	1.69	0.95	20 assumed
Seng 2009	population PK, 59 men	1.96	1.69	0.86	20 assumed
Faber 2004	12 heavy smokers, before vs 13 d after quitting	1.61	1.74	1.08	≥20/day
Joeres 1988	7 smokers, before vs 3 weeks after quitting	1.53	1.69	1.11	20 assumed
Benowitz 2003	12 smokers, smoking vs 7 d of air	1.33	1.64	1.23	20 assumed
Brown 1988	9 smokers, 24-h AUC after 4 d abstinence	1.46	1.51	1.03	AUC ratio; 20 assumed

8.5 What it changes

For 20 cigarettes a day the half-life falls from 5.20 to 3.1 h (3.4 h at 10, 3.9 h at 5 a day). With 150 mg at 8:00 and 14:00, a non-smoker still has 68 mg at 23:00; the same person smoking 20 a day has 26 mg. Two days after quitting the factor is 1.40 (from 1.69), after a week 1.05.

8.6 In the application

- Only for users who track cigarettes; only their own entries (last 28 days), loaded and used in the browser like the caffeine entries. Nothing new is stored, shared or sent.
- The card shows a short line with the current half-life when the factor is above 1.02, and the explanation lists how smoking is used and why snus is not.
- Constants: `smoking` in `ts_constants.json`; `cafSmokingClearance` in `src/lib/caffeinePk.ts` (15-minute grid, exact step for a constant target within a step). `caffeinePk.test.ts` pins the factor and caffeine amounts of the scenario (20 a day for 14 days, then none).

8.7 Limitations of the smoking adjustment

- Individual induction varies more than any population model can capture (genotype, other inducers); the factor is an average for people smoking that many cigarettes.
- Only logged cigarettes count. Unlogged cigarettes, cigars, cannabis smoking (which also induces CYP1A2; Anderson 2016) and heavy coffee drinking (a smaller induction, Tantcheva-Poór 1999; Djordjevic 2008) are not included.
- The onset of induction when starting or resuming smoking is assumed to be as fast as the offset; no human data were found.
- A user who starts logging is assumed to have smoked as in their first week; if they had just started smoking, the first days are overestimated.

9 Deployment specification (for later)

- **Code.** A new `src/lib/caffeinePk.ts` with the constants of `ts_constants.json` (written by `model.py`): `KA_PER_H` = 4.42, `KE_PER_H` = 0.1333, `V_L_PER_KG` = 0.669, `SLEEP_REF_MG_PER_KG` = 0.52, `CUP_MG` = 100; functions for body and gut amounts at a time, a curve for the chart, the amount at bedtime and the time it falls below the reference (closed form for one dose, bisection for several, like `pkTimesBelow` for kratom). Entries in the future (logged for later today) are ignored until their time; entries older than 72 h are ignored.
- **Tests.** `caffeinePk.test.ts` pins the check values in `ts_constants.json` (e.g. 150 mg: 133.5 mg after 1 h, 69.5 mg after 6 h, 31.2 mg after 12 h).
- **Settings.** Body weight (default 70 kg, e.g. 35–250 kg) and bedtime (e.g. 23:00) in *Nastavení* when caffeine is selected; they need two new columns in `user_settings` (one migration), visible only to the owner.
- **UI.** A card on the caffeine *Dnes* page with the mg number, cups line, a chart of the last 24 h plus a projection to the next morning with the bedtime line and the reference, and the honest explanation ("Jak se to počítá?").
- **Privacy.** Computed in the browser from the user's own entries only; nothing new is stored or shared. No notifications are planned.
- **Changing the model** means editing `model.py`, re-running it, copying the constants and re-running the tests, as for kratom.

10 Limitations

- One population half-life (adjusted only for logged cigarettes since version 2); real individuals differ by a factor of three or more (10–90 %: 3.4–8.4 h in the simulated population), most strongly with smoking, oral contraceptives, pregnancy, liver disease and some medicines.
- Linear kinetics; after very large doses or heavy daily intake, elimination may be slower than assumed, so the estimate may then be too low.
- Entries are user-reported; caffeine content of drinks varies widely (a "cup" is not a fixed amount).
- Early peak concentrations are under-predicted; the displayed amount is not affected.
- Tea, cola and chocolate count only if logged. Paraxanthine, which is also active, is not shown.
- The sleep reference is an average effect on total sleep time from studies of mostly young healthy adults and is not a safe limit.

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